

HINT-Blimp: Human INTent Inference from Multimodal Cues for Robotic Blimps

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Abstract—In human-robot shared workspaces, traditional interfaces such as goal-selection menus and handheld tablets introduce latency into navigation tasks and require the operator’s explicit attention. We propose a human-robot interaction framework in which a human instead communicates intent directly through sparse multimodal signals like physical pushes and voice commands. Human intent is represented as a parameterized linear dynamical system (LDS) that encodes the desired goal and motion behavior. The robot estimates this intent (parameters) online using a particle filter, where each particle represents a candidate DS hypothesis and is reweighted online as new information becomes available. We validate this framework on an aerial blimp, whose inherent compliance and collision tolerance make it well-suited for repeated physical interaction. Experiments with five participants show that, in our goal configuration, combining pushes and voice commands achieves 86% goal identification, with the intended goal typically identified within two interactions. The inferred dynamical systems can also produce curved trajectories that avoid obstacles known only to the human. Video: https://youtu.be/P_ZU_f_-qek

I. INTRODUCTION

Understanding human intent is necessary for robots operating in shared environments. During navigation tasks, humans often need to correct or redirect a robot without stopping it, and conventional channels such as goal-selection interfaces introduce latency and demand explicit attention. Physical and verbal cues are more immediate: a brief push, a spoken command, or a pointing gesture can convey a desired goal without an external device. Interpreting such cues is hard because they are short, sparse, and ambiguous. The same push direction may be consistent with several goals or several motion behaviors, especially in clutter or when the desired trajectory is not a straight line. A command like “go left” does not specify a complete trajectory, yet still provides evidence that disambiguates among candidate goals. This motivates a framework that treats each cue not as a complete command, but as partial evidence about the human’s intended goal and trajectory.

Intent inference in physical human-robot interaction has been studied extensively for manipulators in co-manipulation and shared-control tasks [1]–[3], where dynamical systems provide smooth, stable motion representations for physical

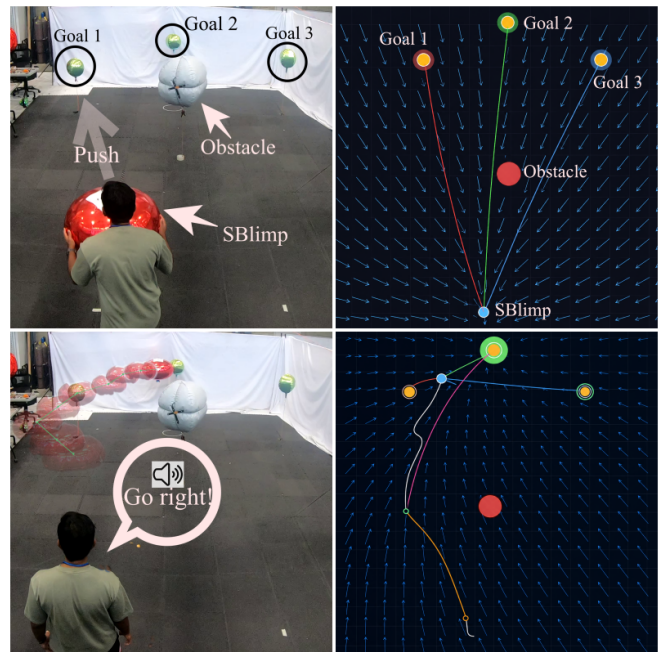


Fig. 1: HINT-Blimp infers human intent from multimodal interactions like physical pushes and voice commands. Here the user intends Goal 2 but cannot push directly toward it, since a straight-line trajectory would hit the obstacle. The user first pushes toward Goal 1, then redirects with “Go right!”. Our method (right column) maintains beliefs over candidate trajectories and converges to the correct intent as more information arrives.

interaction and task adaptation [4]–[6]. Expressing high-level intent through sparse physical cues remains underexplored for mobile robots: prior work on wheeled and aerial platforms has focused on low-level admittance, impedance, or force-estimation control [7]–[11], and on tactile interaction such as virtual touch buttons [12], safe-to-touch interfaces [13], and contact or tether forces [14]–[17]. These treat touch as a compliant control input rather than as evidence for high-level intent.

We address this gap using aerial blimps, whose low-speed dynamics, compliance, and collision tolerance suit them to repeated physical interaction [18]. Our contribution is a multimodal intent-inference framework in which intent is represented jointly by a candidate goal and the parameters of a dynamical-system motion policy, so that cues inform both *where* the human wants the robot to go and *how* it should get there. Pushes, spoken commands, and pointing

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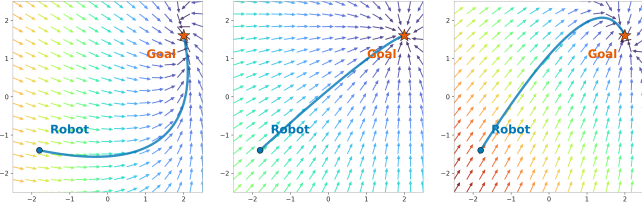


Fig. 2: Vector fields and robot trajectories for LDS. From left to right, the parameters are $(r, \theta_1, \theta_2) = (0.58, 103^\circ, 133^\circ)$, $(0.90, 117^\circ, 70^\circ)$, and $(0.47, 117^\circ, 70^\circ)$.

gestures enter a unified Bayesian estimation problem, and the belief is maintained online with a particle filter that preserves uncertainty over goals and motion behaviors. We validate the framework on a physical blimp with five participants.

II. PROBLEM FORMULATION

A human wishes to send a robotic blimp to one of K candidate goals $\{\mathbf{x}_i^*\}_{i=1}^K$, communicating this through sparse multimodal interactions. We write $\mathbf{x} \in \mathbb{R}^2$ and $\psi_{\text{robot}} \in S^1$ for the blimp’s position and heading. Intent is modeled as the LDS

$$\dot{\mathbf{x}} = \mathbf{A}(\mathbf{x} - \mathbf{x}^*), \quad (1)$$

where $\mathbf{A} \in \mathbb{R}^{2 \times 2}$ has real, strictly negative eigenvalues, guaranteeing global asymptotic stability at $\mathbf{x}^*$. The eigenstructure of $\mathbf{A}$ sets the qualitative motion: isotropic contraction gives straight-line convergence, while unequal contraction rates along the eigendirections give curved trajectories (Fig. 2). We restrict $\mathbf{A}$ to the stable-node case, and infer $\theta = (\mathbf{x}^*, \mathbf{A})$.

The robot receives three inputs. **Scene context** $\mathcal{S} = \{(\mathbf{x}_i^*, \pi_i)\}_{i=1}^K$ supplies candidate goals with prior plausibilities π_i ; unlike the others it is not a human interaction, but background knowledge about which destinations matter. **Physical pushes** are short bursts of motion imparted by the human, each recorded as a trajectory segment $\mathcal{D}_\tau^{(j)} = \{\mathbf{x}(t), \dot{\mathbf{x}}(t)\}$ over a window of length τ . **Directional hints** let the human redirect the blimp without contact, by saying “go left” or pointing. Treating speech and gesture as two sources of the same evidence, each yields a payload

$$\mathbf{u}_j = (\mathbf{d}_j^{\text{cmd}}, \mathbf{x}_{\text{robot},j}, \psi_{\text{robot},j}) \quad (2)$$

recording a commanded direction $\mathbf{d}_j^{\text{cmd}} \in S^1$ in the robot’s heading frame, with the blimp’s position and heading at command time. These arrive as an ordered sequence $\mathbf{z}_{1:k}$ in any order and at any time, and the robot must infer from it the $\mathbf{x}^*$ and $\mathbf{A}$ defining the DS it should then follow.

III. METHOD

Eigenvalue parameterization. Rather than estimating the entries of $\mathbf{A}$, we parameterize it through its eigen-decomposition. Requiring a stable node, we fix $\lambda_1 = -1$ and parameterize the relative contraction rate $r = |\lambda_1|/|\lambda_2| \in (0, 1]$, giving

$$\mathbf{A} = \mathbf{V} \text{diag}(-1, -1/r) \mathbf{V}^{-1}, \quad (3)$$

where the columns of $\mathbf{V}$ are unit eigenvectors at angles $\vartheta_1, \vartheta_2 \in [0, \pi)$, kept minimally separated so $\mathbf{V}$ stays invertible. Each particle is thus $\theta = (\vartheta_1, \vartheta_2, r, \mathbf{x}^*)$. This gives us a lower-dimensional, structured search space that enforces stability by construction.

Particle filter. We represent the belief over intent as N weighted particles $\{\theta_k^{(n)}, w_k^{(n)}\}_{n=1}^N$, each a candidate DS hypothesis, and follow the standard predict-weight-resample cycle [19]. Particles are initialized by sampling goals in proportion to π_i . Since intent does not change between interactions, the process model is the near-static walk $\theta_k^{(n)} = \theta_{k-1}^{(n)} + \eta_k^{(n)}$, with $\eta_k^{(n)}$ zero-mean Gaussian noise on the shape components $(\vartheta_1, \vartheta_2, r)$ only. Goals are a discrete hypothesis set held fixed at initialization, so the belief over them evolves only through reweighting and resampling.

Push likelihood. Pushes give direct kinematic evidence about the intended goal. The j -th push is scored against particle n by the mean cosine similarity between observed velocities and those predicted by that particle’s DS, $\hat{\mathbf{x}}_t^{(n)} = \mathbf{A}^{(n)}(\mathbf{x}_t - \mathbf{x}^{*(n)})$:

$$p(\mathcal{D}_\tau^{(j)} | \theta^{(n)}) \propto \exp(\kappa_p \bar{c}_j^{(n)}), \quad \bar{c}_j^{(n)} = \frac{1}{N_\tau} \sum_t \frac{\dot{\mathbf{x}}_t \cdot \hat{\mathbf{x}}_t^{(n)}}{\|\dot{\mathbf{x}}_t\| \|\hat{\mathbf{x}}_t^{(n)}\|}, \quad (4)$$

with $\kappa_p > 0$ and N_τ the number of samples in the window. This directional scoring is consistent with the speed-invariant parameterization of the DS. Only pushes carry evidence about $(\vartheta_1, \vartheta_2, r)$, so the process noise is gated on push alignment: it is frozen until the first push, then scaled down as the push history becomes better explained by the current estimate.

Directional likelihood. A direction command is evidence about where the goal lies: particles whose goals sit along the commanded direction should gain weight, others should lose it. We measure both angles in the robot frame, $\alpha_j^{(n)}$ toward particle n ’s goal, and $\alpha_j^{\text{cmd}} = \text{atan2}(d_{y,j}^{\text{cmd}}, d_{x,j}^{\text{cmd}})$ from the payload $\mathbf{u}_j$ and take their difference $\Delta\alpha_j^{(n)}$ wrapped to $[-\pi, \pi]$:

$$p(\mathbf{z}_j^{\text{dir}} | \theta^{(n)}) \propto \kappa_d^{\gamma_j^{(n)}}, \quad \gamma_j^{(n)} = \max\left(e^{-(\Delta\alpha_j^{(n)})^2 / 2\sigma_d^2}, \epsilon_d\right), \quad (5)$$

with $\kappa_d > 1$. A perfectly aligned goal receives the full boost κ_d , which decays as a Gaussian of width σ_d with misalignment. The floor $\epsilon_d > 0$ keeps misaligned particles down-weighted rather than eliminated, so the true goal can recover from a misspoken command or a biased heading estimate. Because $\mathbf{d}_j^{\text{cmd}}$ enters only as a unit vector, the formulation is not tied to a fixed vocabulary; we use four commands, straight, right, left, back, mapped to $(1, 0)$, $(0, 1)$, $(0, -1)$, $(-1, 0)$.

Resampling and commitment. Systematic resampling runs at every update carrying evidence. This is also how evidence accumulates: the particle set entering an update carries uniform weights, so the posterior is encoded by particle density in parameter space rather than by weights. The filter commits once an observation is complete, after the window

$\tau = 5$ s for a push, at the decoding step for a hint. Since goals are static, particles partition into one cluster $\mathcal{N}_i$ per goal, and we commit to the cluster with the largest total weight, $i_k^* = \arg \max_i \sum_{n \in \mathcal{N}_i} w_k^{(n)}$: cluster population records past evidence, weights record the current cue. The shape comes from the highest-weight particle in that cluster, giving the policy $\dot{\mathbf{x}} = \mathbf{A}^*(\mathbf{x} - \mathbf{x}^*)$ with $\mathbf{A}^*$ from (3), followed until a new interaction arrives.

IV. EXPERIMENTS AND RESULTS

We use an SBlimp [18], a Crazyflie 2.1 with tilted rotors attached to a Mylar helium balloon. Its pendulum-like stability lets us command translational velocity directly without an attitude loop, which is exactly the interface the inferred LDS produces: $\mathbf{A}^*(\mathbf{x} - \mathbf{x}^*)$ is evaluated at the current position and sent as a desired velocity to the onboard controller. The blimp is localized with OptiTrack, and commands issued through crazyswarm [20]. The filter runs in NumPy on CPU. Pushes are detected when motion-capture speed exceeds a threshold inconsistent with the commanded velocity; voice is decoded with Vosk [21] and gestures with MediaPipe Hands [22], both in background threads. Only the velocity controller runs on the blimp.

Across all scenarios the robot has no knowledge of obstacles, while the human knows the layout and interacts consistently with avoiding them. For Experiments 1 and 2 the scene prior is uniform, so the belief is shaped entirely by human interaction. A trial is a *success* if the robot reaches the predesignated goal in fewer than 6 interactions. All reported trials use pushes and voice only, the gesture channel is implemented but not evaluated separately, since gestures and voice reduce to the same payload and likelihood.

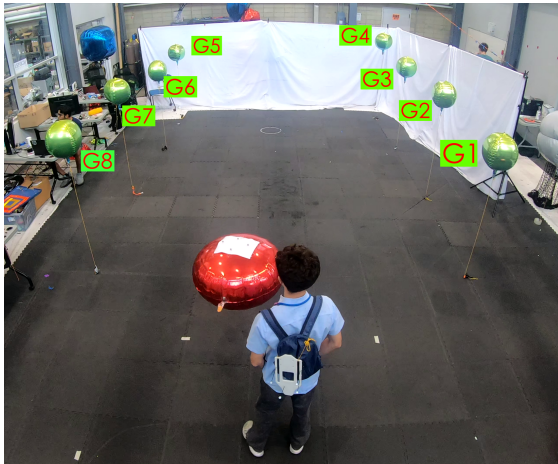


Fig. 3: Experiment 1 setup: the user sends the robot to one of 8 candidate goals, selected randomly at the start of each run.

A. Experiment 1: Goal Identification and Modality

Five participants each completed 60 trials (180 total) in a scene with $K = 8$ candidate goals (Fig. 3), one selected uniformly at random per trial and unknown to the robot.

TABLE I: Success rate by interaction modality. Per-participant cells are successful trials out of 20; the pooled column is out of 100. SD is the sample standard deviation of the five participant rates.

Modality	Participant					Pooled	
	1	2	3	4	5	n	SD
Push only	14	18	16	15	13	76	9.6
Voice only	12	11	14	15	11	63	9.1
Combined	15	18	18	17	18	86	6.5

Participants conveyed the target under three conditions: push only, voice only, or combined.

Table I and Fig. 4 show the three modalities succeeding through different interaction patterns. Push-only is a high-information channel: over half its successes resolve in a single interaction, but when the first push is ambiguous the participant must physically return to the robot to correct it, increasing effort. Voice-only gives weaker directional evidence per interaction, spreading successes across one to five commands and producing the highest failure rate (37/100). The combined condition resolves this, successes concentrate sharply at two interactions, consistent with a two-stage strategy where a push sets coarse direction and a voice command disambiguates among the remaining candidates.

Participants rated ease of conveying their goal and confidence in the robot’s understanding on a 5-point Likert scale. All five rated both at 4 or above, and four preferred the combined modality. All independently reported a “push-then-voice” strategy, matching the two-interaction peak in Fig. 4. Participants identified the central targets, particularly Goals 3 and 6, as hardest to communicate because of their proximity to neighbors.

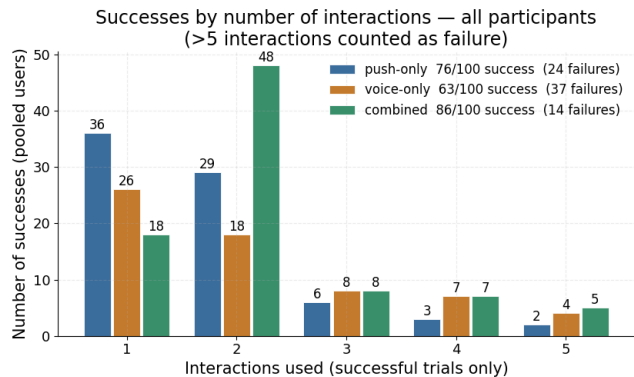


Fig. 4: Distribution of the number of interactions used in successful trials, pooled across all five participants.

B. Experiment 2: Curved Trajectories Around Obstacles

Because the eigenvalue parameterization admits any stable node, the inferred DS is not restricted to straight-line convergence, letting the human steer the robot around obstacles known only to them without ever communicating the obstacle location. We tested this with 3 goals and a single obstacle,

tasking the participant with guiding the robot to a goal whose straight-line path was blocked.

Fig. 1 shows a case where the human wants Goal 2 but a straight line would collide with the obstacle. Rather than pushing toward Goal 2, the human pushes toward Goal 1. After this push the filter retains hypotheses for all goals, with Goal 1 weighted most heavily since the push aligns with the straight-line direction to it. The human then says “go right,” reweighting particles whose goals lie right of the robot’s heading. The filter commits to Goal 2 and the executed trajectory curves around the obstacle.

C. Experiment 3: Sensitivity to Priors

We repeated the combined condition under three initializations: uniform, helpful (40% of particles on the true target), and adversarial (40% on the goal farthest from the true target). To isolate the prior from sensor and actuation noise, these ran in simulation with a similar filter configuration; absolute rates may differ in the real world. Each condition comprised $n = 40$ trials (Fig. 5).

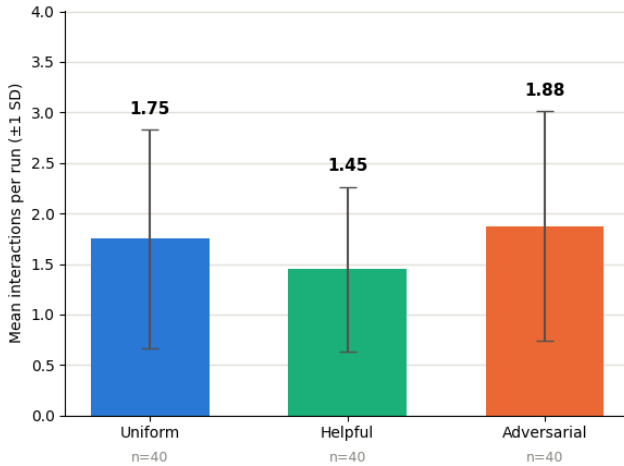


Fig. 5: Interactions to successful commitment under three prior conditions ($n = 40$ each). The adversarial prior matches the uniform baseline despite concentrating 40% of particles on the farthest goal; the helpful prior shows a modest reduction.

The framework is robust to priors. Even with 40% of its mass on the farthest goal, the adversarial prior needed about as many interactions as the uniform baseline (1.88 vs. 1.75), showing one informative cue is enough to overcome a bad initialization. The filter is evidence-dominated: resampling at every step lets the first cue reshape the particle cloud quickly and largely overwrite the prior. The helpful prior did somewhat better (1.45), but the effect is small and we do not claim it as a strong result.

V. CONCLUSION

We presented a multimodal intent-inference framework representing human intent as a linear dynamical system whose parameters are estimated online with a particle filter.

Pushes, voice commands, and gestures enter as complementary evidence through modality-specific likelihoods, letting the robot accumulate partial cues rather than requiring any single interaction to fully specify the goal. On a physical blimp, combining pushes and voice achieved 86% goal identification with a two-interaction convergence pattern, outperforming either modality alone, and the eigenvalue parameterization let the filter infer curved trajectories avoiding obstacles known only to the human.

Three limitations bound these results. Five participants is too small a sample to characterize variation in interaction strategy, so the reported preferences should be read as qualitative. The pointing-gesture decoder is implemented but not evaluated in isolation; since it produces the same payload and likelihood as a voice command, its reliability depends on the hand-pose estimator rather than the filter, and remains unmeasured. Finally, the robot has no obstacle awareness of its own, avoidance emerges entirely from the inferred DS, so a mis-decoded cue can commit it to a colliding trajectory with no independent check. Future work will scale the study, evaluate the gesture decoder separately add a reactive safety layer that modulates the DS output against onboard range sensing.

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